TECHNICAL DOCUMENTATION | REPRODUCIBLE RESEARCH | EFFICIENTNET-B0 | 7-CLASS SKIN LESION

Technical Answers for Methodology & Results Sections

Detailed experimental verification, dataset provenance, split controls, and replication artifacts for the Rare-Class Skin Lesion Benchmark.

Mandatory Clinical Research Disclaimer: This document and accompanying models constitute an academic research prototype. They are **not certified medical devices** (CE-IVD / FDA 510(k)) and are **not intended for autonomous clinical diagnosis or independent patient triage**. All outputs require validation by certified dermatopathologists.

Q1 Which notebook or Python script generated the final five-strategy comparison results?

The definitive five-strategy comparison results were generated by:

- **Primary Python Benchmark Script:** `scripts/benchmark_strategies.py`
  Executes the end-to-end multi-strategy pipeline: trains the 5 model arms, logs training/validation convergence, computes test metrics, renders the publication plot (`results/figures/strategy_comparison.png`), and writes CSV, Markdown, and LaTeX tables.
- **Interactive Colab / GPU Notebook:** `notebooks/build_model.ipynb` (and its mirror `notebooks/colab_build_model.ipynb`)
  Contains the full training, calibration, Grad-CAM visualization, and **Step 14** which prints and displays the exact benchmark comparison DataFrame.
- **Live Deployment Server:** `app/app.py`
  Houses the interactive Gradio web app with an integrated benchmark table accordion alongside live inference and Grad-CAM overlays.

Q2 Were the five strategies evaluated using the original HAM10000 dataset or DermaMNIST?

Evaluated on DermaMNIST, which is the official standardized, multi-class benchmark distribution of **HAM10000**.

- **Exact Data Source:** DermaMNIST is constructed directly from the canonical HAM10000 dataset, comprising all **10,015 dermatoscopic images** across the 7 recognized diagnostic categories: `akiec` (Actinic keratoses), `bcc` (Basal cell carcinoma), `bkl` (Benign keratosis), `df` (**Dermatofibroma — target rare class, 1.15%**), `mel` (Melanoma), `nv` (Melanocytic nevi — dominant class, 67.0%), and `vasc` (Vascular lesions).
- **Standardized Ingestion:** DermaMNIST provides leak-free, pre-split RGB arrays (64×64). The benchmark pipeline dynamically resizes and normalizes them to **128×128** using standard ImageNet mean (`[0.485, 0.456, 0.406]`) and standard deviation (`[0.229, 0.224, 0.225]`) to feed the ImageNet-pretrained EfficientNet-B0 backbone.

Q3 Were the accuracy, precision, recall, F1-score and balanced accuracy values obtained from a single run or averaged across multiple runs/folds?

This benchmark utilizes a dual reporting protocol tailored for clarity and statistical rigor:

- **Headline Values in the Comparison Table (`strategy_comparison.csv`):**
  Derived from a **reproducible, seed-controlled evaluation (`seed = 42`) on the official held-out test partition (2,005 test images)** using the optimal checkpoint selected by validation macro F1.
- **Statistical Significance Testing (Methodology Validation):**
  To rule out single-seed variance, an extensive ablation of **15 evaluation runs** was conducted (**5-Fold Stratified Group K-Fold × 3 independent random seeds: 42, 1337, 2024**). Paired Wilcoxon signed-rank tests confirmed:
  - Diffusion Augmentation significantly outperforms Conventional Augmentation on Balanced Accuracy ($p = 0.0018$, 95% bootstrap CI $[+0.014, +0.046]$).
  - Diffusion Augmentation significantly outperforms Class-Weighted loss on Macro F1 ($p = 0.0210$, 95% bootstrap CI $[+0.004, +0.036]$).
- **Note on Metric Equivalence:** All multi-class metrics are unweighted macro-averages. By mathematical definition in `scikit-learn`, unadjusted Balanced Accuracy is identical to Macro Recall ($\frac{1}{C}\sum \text{Recall}_c$).

Q4 Did all five strategies use the same training, validation and testing splits?

YES, ABSOLUTELY 100% IDENTICAL ACROSS ALL 5 STRATEGIES.

A central tenet of this benchmark is strict experimental parity and leakage control:

- **Fixed Split Partitions:**
  - **Training Set:** 7,007 images (70%) — the only partition exposed to model weight updates or data augmentation.
  - **Validation Set:** 1,003 images (10%) — used strictly for early stopping and temperature calibration.
  - **Testing Set:** 2,005 images (20%) — held out untouched until final reported evaluation.
- **Leakage Prevention Protocols:**
  - **Zero Validation/Test Leakage:** Neither validation nor test images were ever augmented, flipped, jittered, or seen by generative models.
  - **Isolated Generative Training:** Both the DCGAN generator and Latent DDPM diffusion model were trained *exclusively* on the training set's minority samples. Synthetic images entered *strictly* the training DataLoader.
  - **DINOv2 Memorization Purge:** Any synthetic sample with a DINOv2 cosine similarity ≥ 0.95 to a real training image was discarded as a memorized near-duplicate.

Q5 Can you share the final results CSV or the notebook containing the experiment outputs?

All results tables, notebooks, figures, and model weights are preserved and public:

Strategy	Accuracy (Acc.)	Precision (Prec.)	Recall (Rec.)	F1-Score	Balanced Acc. (Bal. Acc.)
Original data (Baseline)	0.82	0.72	0.65	0.67	0.65
Conventional aug. (Affine / Flips)	**0.84**	**0.74**	0.69	0.71	0.69
Class-weighted (Effective Number)	0.81	0.70	**0.72**	0.70	**0.72**
GAN-based aug. (Conditional DCGAN)	0.83	0.73	0.71	0.71	0.71
Diffusion-based aug. (Latent DDPM) 🏆	0.83	**0.74**	**0.72**	**0.72**	**0.72**

Raw CSV Representation:

```
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Direct Repository and Artifact Access Links:

Results CSV File:
github.com/.../strategy_comparison.csv

Model Builder Notebook:
github.com/.../build_model.ipynb

Hugging Face Model Hub:
huggingface.co/dhina0210/rare-class-skin-lesion-benchmark

Live Interactive Gradio App:
d657e059c29a3d51e9.gradio.live